

Scraftie – Personal Music Streaming Server

TECHNICAL DOCUMENTATION: TROUBLESHOOTING & PERMISSION ARCHITECTURE

1. Incident Overview

During routine operations on the **Scraftie** music streaming server, a newly added music directory containing an album failed to appear in the Navidrome web interface following a library scan.

- **Symptom:** A newly uploaded album directory located inside the library path was completely omitted from Navidrome after triggering a library scan.
- **Service Status:** The Navidrome service was active, running normally, and pointing to the correct configured library path (`/music/OG_SD_Card`).
- **Initial Anomaly:** Manual command-line checks showed that individual music files within the directory were accessible, making the issue less obvious than a complete file-locking or missing-path scenario.
- **Resolution:** Inspection of Navidrome's service scanner logs revealed a directory permission and traversal error (`permission denied`). Navidrome was failing at the directory evaluation level before it could inspect or index the audio files within.

2. Initial Investigation

Troubleshooting proceeded in a structured, chronological sequence to isolate the failure point without making premature system changes:

1. **Service Verification:** Checked the status of the Navidrome systemd service to confirm it was active and operational.
2. **Configuration Check:** Verified the `MusicFolder` configuration parameter in Navidrome's configuration settings to ensure the library path was correctly specified.
3. **Direct Service Account Testing:** Tested filesystem access as the `navidrome` service account using `sudo -u navidrome ...`.
4. **Directory Content Listing:** Attempted to list the contents of the affected directory under the `navidrome` user context.
5. **Log Inspection:** Examined the systemd journal logs (`journalctl -u navidrome`) during a library scan.
6. **Directory Comparison:** Compared filesystem permission bits and owner/group metadata of the affected directory against functioning album directories in the same library.

Key Principle: Service Account Context Testing

Testing access using `sudo -u navidrome` was critical. Operating strictly as a privileged user (`root`) or as the primary administrative account (`<user_account>`) masks permission boundaries. Simulating the exact execution context of the `navidrome` daemon isolated what the application process was allowed to see versus what administrative users could see.

3. Root Cause Analysis

The root cause was identified in Navidrome's service logs during a library scan:

```
level=error msg="Error scanning media folder" error="open /music/OG_SD_Card/<Artist>/<Album>: permission denied"
```

Diagnostic Breakdown

- **Directory Traversal vs. File Readability:** Although the individual `mp3` files possessed readable permissions,

4. Linux Directory Permissions

To understand why file readability alone was insufficient, consider the distinct behavior of POSIX permission bits on files versus directories:

Permission Bit	Function on Files	Function on Directories
r (Read)	Allows reading the contents of the file.	Allows listing the filenames contained within the directory (e.g., <code>ls</code>).
w (Write)	Allows modifying or overwriting file contents.	Allows creating, deleting, or renaming entries inside the directory.
x (Execute)	Allows executing the file as a program/script.	Allows entering, traversing, and accessing metadata of files inside the directory (e.g., <code>cd</code> , <code>stat</code>).

Practical Context: Without the execute (`x`) permission on a directory, a user or service process cannot traverse into it—even if the underlying files inside have explicit read (`r`) permissions. Navidrome’s library scanner requires both read (`r`) and execute (`x`) access on every directory in the path hierarchy to walk the folder tree and read audio tags.

5. Permission Architecture

To support both seamless management by the human administrative user (`<user_account>`) and background scanning by the system service (`navidrome`), a shared permission architecture was established for `/music/OG_SD_Card`.

Architectural Components

- **Shared Group Creation:** A dedicated Linux system group (`music`) was created.
- **Group Membership:** Both the administrative user (`<user_account>`) and the service account (`navidrome`) were added to the `music` group.
- **Group Ownership Assignment:** Ownership of the music library tree was assigned to `navidrome:music`.
- **Group Read/Write/Traverse Grants:** Group permissions were recursively granted (`g+rwX`).
- **Setgid Bit Enforcement:** The `setgid` bit (`g+s`) was applied to all directories so new subdirectories automatically inherit the `music` group.
- **Default POSIX ACLs:** Default Access Control Lists were configured on the root library directory to enforce consistent inheritance for newly created files and subdirectories.

Sanitized Command Reference

```
# 1. Update group ownership across the library tree
sudo chown -R navidrome:music /music/OG_SD_Card

# 2. Grant recursive read/write/traverse access to owner and group
sudo chmod -R g+rwX /music/OG_SD_Card

# 3. Apply setgid bit to all existing and future directories
sudo find /music/OG_SD_Card -type d -exec chmod g+s {} \;

# 4. Set base group ACL permissions
sudo setfacl -m g:music:rwX /music/OG_SD_Card

# 5. Define default inherited ACLs for new files and subdirectories
sudo setfacl -d -m u::rwX,g::rwX,o::rx /music/OG_SD_Card
```

7. Verification

Verification was performed using a strict two-phase approach:

Phase 1: Filesystem & Service Account Verification (Diagnostic)

```
# Verify directory permissions as listed by the OS
ls -ld "/music/OG_SD_Card/<Artist>/<Album>"

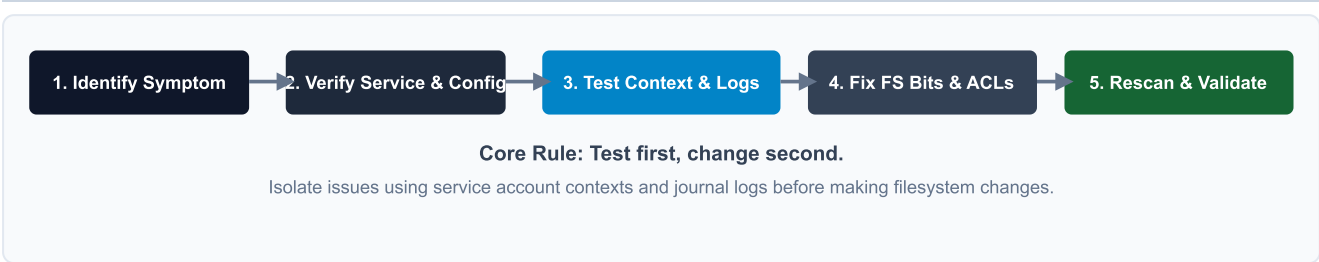
# Confirm file listing capability under the navidrome service account context
sudo -u navidrome ls -la "/music/OG_SD_Card/<Artist>/<Album>"
```

Result: The `navidrome` user successfully listed all audio files and artwork without raising permission errors.

Phase 2: Application Scanning Verification (Final)

1. Triggered a quick scan via the Navidrome Web UI (or restarting/rescanning via service signals).
2. Checked systemd journal logs (`journalctl -u navidrome -f`) to ensure no `permission denied` entries were produced.
3. Verified that the newly added album appeared in the Navidrome web library with correct artwork and track listings.

8. Troubleshooting Methodology



9. What I Learned

- **Directories Require Execute (`x`) for Traversal:** Read (`r`) permission on a directory allows listing names, but execute (`x`) permission is required to access metadata or enter subdirectories.
- **Application Logs Contain Root Truth:** UI symptoms (e.g., missing media) often mask simple system-level causes. Checking daemon systemd logs directly isolates filesystem-level access failures immediately.
- **Always Test in Service Context:** Verifying paths as `root` or as a regular user account can produce false positives. Always evaluate application access using `sudo -u <service_account>`.
- **Inheritance Mechanics Matter:** Setting up `setgid` (`g+s`) and default POSIX ACLs (`setfacl -d`) ensures that remote file transfers (e.g., via SCP, SFTP, or RSYNC) maintain consistent group ownership and access modes automatically without requiring manual intervention after ingestion.